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Flooding in Sacramento county, California, on 1 January

underused canal into wells. He says a single well can add around 1 million litres of water per day to the aquifer below it during the right conditions. Other projects flood fields outside the growing season or flood crops that can tolerate the water.

In the San Francisco Bay Area, the Alameda County Water District maintains inflatable rubber dams that can divert stormwater flowing in creeks into gravel quarries, where it seeps down into the groundwater basin.

"We're sipping off of those big flows," says Fisher, who has a test project diverting stormwater running off hillsides into a basin in the Pajaro valley near Santa Cruz. "We're taking a fraction of that water, we're slowing it down and we're getting it to locations where it can get into the ground."

There are growing efforts to capture more stormwater in urban areas as well. Luthy points to two parks under construction in Los Angeles and San Francisco that double as aquifer recharge sites. A report from the Los Angeles Department of Water and Power found that the city could feasibly capture enough stormwater to supply roughly a quarter of its water by 2035.

There are many obstacles to expanding recharge projects, from securing water rights and land to dealing with contamination, which can be a problem with runoff from both urban and agricultural land. But "we're looking at stormwater differently in the 21st century", says Luthy.

In all, 340 local recharge projects have been proposed in the state, according to a report by Californian state water agencies. If all of these were built, they could store around 570 billion litres of water in a wet year. ■

Environment

Capturing the storm

Heavy rain in California continues to harm the state, but plans to store rainwater could help stave off drought, reports **James Dinneen**

SWATHES of California have seen "atmospheric rivers" dump heavy rains since the end of December, with storms forecast to bring trillions more litres in January.

The deluge has wreaked havoc and led to at least 12 deaths, but it could also offer some respite to a state in the throes of a severe water crisis.

While some of the stormwater will top up depleted reservoirs and aquifers, much of it will flow off pavements and farmland into rivers and drainage systems to be carried out to sea. Several projects are testing efforts to capture water before it is lost.

"Historically, we've viewed stormwater as a nuisance," says Andrew Fisher at the University

of California, Santa Cruz. "The approach that a lot of places have taken is to get it off the landscape as quickly as possible."

The water crisis in the western US is changing that thinking. Across California, water managers and researchers are focusing on "managed aquifer recharge" to capture as much stormwater as possible underground.

The approach has been around for decades, but it wasn't a priority when water could be imported from elsewhere and deficits could be made up by pumping groundwater, says Richard Luthy at Stanford University in California. "The day of reckoning hadn't come yet."

It has now. Climate change has

reduced snowpack – the state's largest store of water – and a drying Colorado river offers less water to import. A recent study found that California's heavily irrigated Central valley has lost around 2 cubic kilometres of groundwater to unsustainable pumping each year since 1961.

570bn

Capacity of proposed Californian water recharge projects, in litres

The crisis has spurred a hunt for other sources of water. "People are serious about this," says Rob Gailey, a water consultant working on a project in Central valley to pump stormwater from an